

A PROJECT TITLE
ON
Credit Card Fraud Detection Using Logistic Regression and Isolation-Forest Anomaly Detection

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
A.Y.: 2026-27

by

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DEPARTMENT OF

Artificial Intelligence & Data Science and Computer Science & Information Technology

Name of Title: Credit Card Fraud Detection Using Logistic Regression and Isolation-Forest Anomaly Detection

1. Abstract:

Credit Card Fraud Detection Using Logistic Regression and Isolation Forest Anomaly Detection is a machine learning-based project designed to identify fraudulent credit card transactions with high accuracy and efficiency. As the use of digital payment systems continues to grow worldwide, credit card fraud has become one of the most significant challenges faced by financial institutions and customers. Traditional fraud detection methods often struggle to detect new and evolving fraud patterns, making intelligent, data-driven solutions increasingly important.

This project combines two powerful machine learning techniques: Logistic Regression, a supervised learning algorithm, and Isolation Forest, an unsupervised anomaly detection algorithm. Logistic Regression is used to classify transactions as either legitimate or fraudulent based on previously labeled transaction data. It learns the relationship between transaction features and fraud labels, enabling accurate prediction of known fraud patterns. On the other hand, Isolation Forest identifies unusual or anomalous transactions by isolating data points that differ significantly from normal behavior. This allows the system to detect previously unseen or emerging fraudulent activities without requiring labeled training data.

The project involves several stages, including data preprocessing, feature scaling, handling class imbalance, model training, testing, and performance evaluation. Standard evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are used to measure the effectiveness of both models. A comparative analysis is also performed to understand the strengths and limitations of each approach in detecting fraudulent transactions.

The proposed system aims to improve the reliability and speed of fraud detection while minimizing false positives and false negatives. By integrating supervised classification with anomaly detection, the solution provides a more comprehensive and adaptive approach to identifying suspicious financial transactions. This project demonstrates how machine learning techniques can enhance financial security, reduce economic losses, and support real-time fraud prevention. The developed model has the potential to assist banks, financial organizations, and online payment platforms in making faster, more accurate decisions, ultimately improving customer trust and the overall safety of digital payment ecosystems.

2. **INTRODUCTION:**

The increasing use of online banking and digital payment systems has led to a rise in credit card fraud, making accurate fraud detection essential. Traditional methods often fail to identify new and evolving fraud patterns efficiently.

This project, "**Credit Card Fraud Detection Using Logistic Regression and Isolation Forest Anomaly Detection,**" uses machine learning to detect fraudulent transactions. Logistic Regression classifies transactions based on labeled data, while Isolation Forest identifies unusual or suspicious transactions by detecting anomalies.

The system includes data preprocessing, model training, testing, and performance evaluation using metrics such as Accuracy, Precision, Recall, and F1-Score. The proposed solution helps improve fraud detection accuracy, reduces financial losses, and enhances the security of digital payment systems.

3. **SOFTWARE REQUIREMENTS:**

The following software tools and technologies are required to develop and implement the Credit Card Fraud Detection system:

Software	Purpose
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.10 or above
Development Environment	Jupyter Notebook or Visual Studio Code
Machine Learning Library	Scikit-learn
Data Processing	Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Dataset	Credit Card Fraud Detection Dataset (CSV)

Python Libraries

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Joblib (for saving trained models)

Minimum Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB (8 GB Recommended)
- Storage: 500 MB free disk space
- Internet Connection: Required for downloading datasets and libraries

Signature of the Guide

Course Instructor